



ISE 726: Systems Modeling

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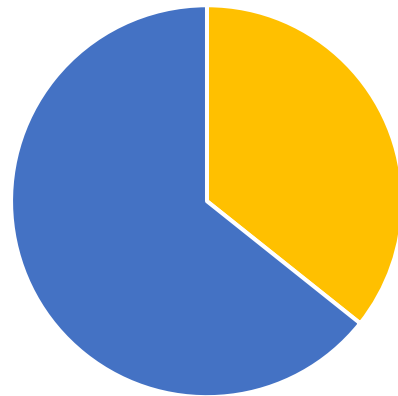
- To comply with federal Title IV financial aid requirements, UAH must verify that students have begun attending or academically participating in each course for which they are registered.
- For this reason, all students must demonstrate participation during the first **10 days of class**. Participation may include attending class, submitting an assignment, completing a quiz or exam, or participating in another academically related activity.
- Please note that simply being registered for the course does not count as attendance. If a student does not begin attending or participating, this may affect financial aid eligibility.

Learning Objectives

- **Introductions**
 - Personal Introductions
 - Course Introduction, incl. Learning Objectives
- Systems Engineering (SE) Overview
 - A bit of history
 - The “V”-Model
 - SE Standards and Handbooks
 - Challenges of current practice
- Stakeholder Analysis
 - Identifying Stakeholders
 - CONOPS
 - Stakeholder Value Network (SVN) Analysis

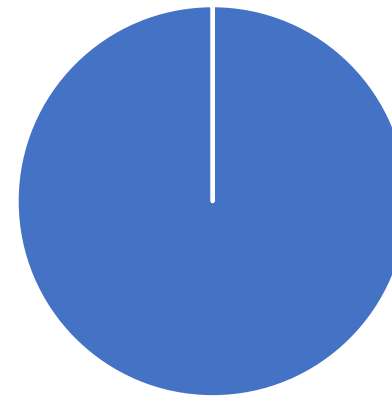
Grades Distributions:

9 – As and 5 - Bs



■ F ■ D ■ C ■ B ■ A

25 A(s)



■ F ■ D ■ C ■ B ■ A

Reciprocal Interview:

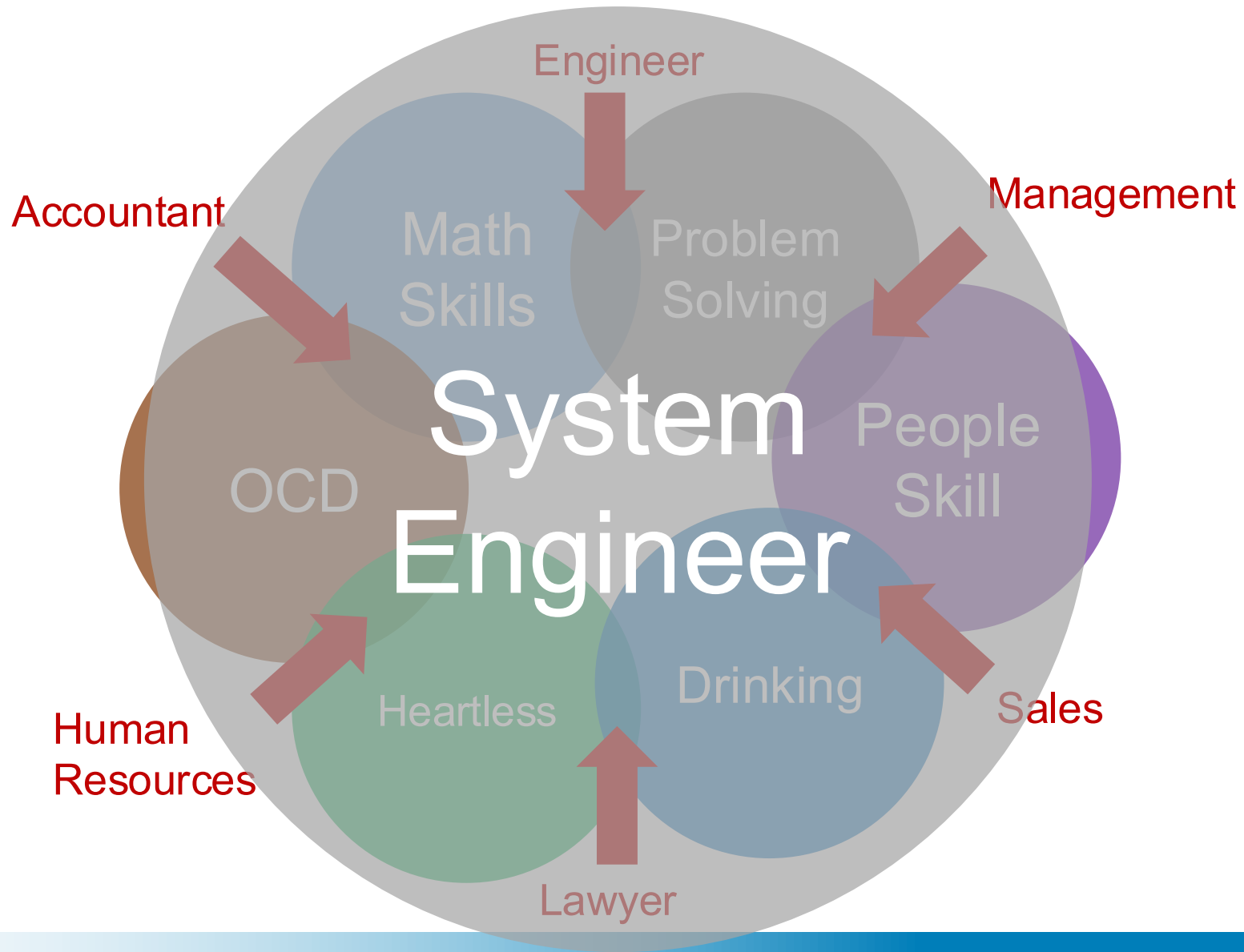
What excited you about this class?

What are your goals for this course?

What can I do to help you throughout this semester?

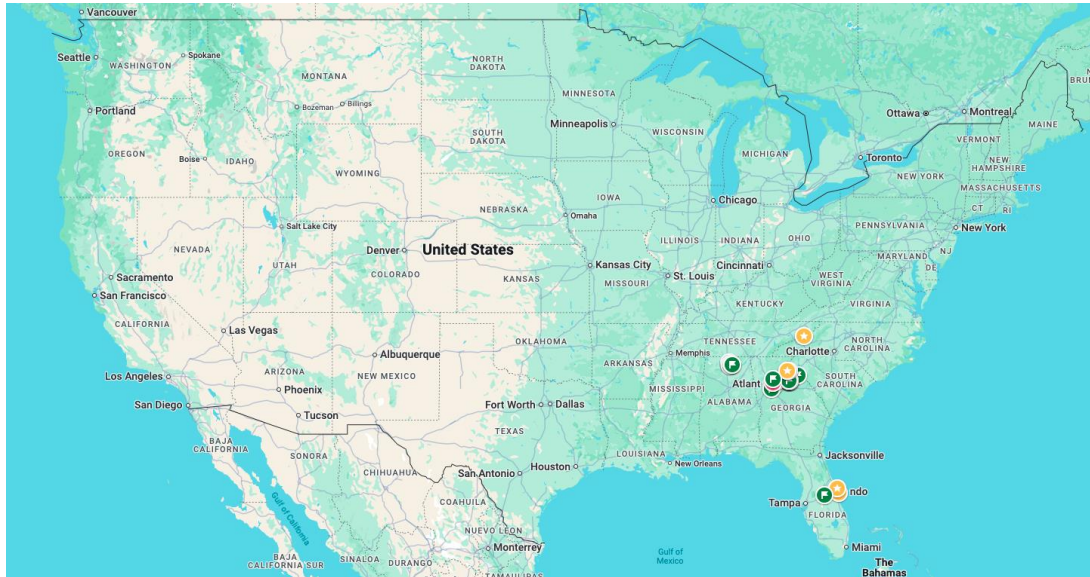
What frightens you about this class?

Why become an engineer? – Venn diagram



About the instructor:

Where I've lived:



Hobbies:



Education:



Ph.D. in Engineering



M.S., in Aerospace Engineering



B.S., in Mechanical Engineering



Self-Introduction

- Your name, department/lab affiliation, and research interests
- Any prior experience with system engineering? Any programming experience?
- Name one thing you want to learn in this class

Let's review the syllabus together.

Email Etiquette:

- When email a teacher, ALWAYS include your **full name and course**
- Include your class and what the email is specifically regarding in the **subject**
- Example:

Subject: ISE 430: Question about Homework 3

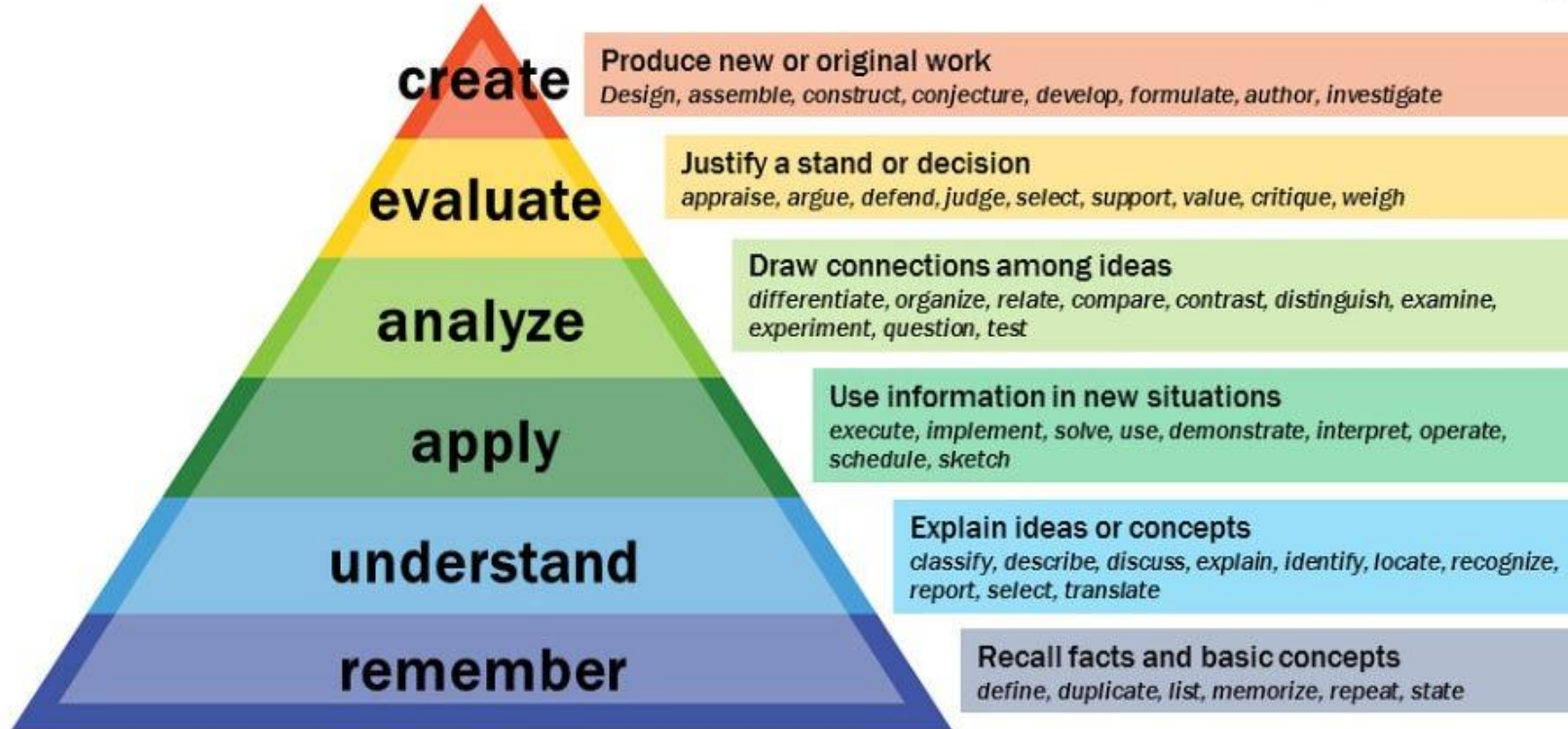
Additional Notes:

- Unless otherwise instructed, each student can collaborate on assignments **together**. However, discussion (not copying) of assignments with classmates is encouraged.
- Students are expected to keep the neatness and professionalism on all items submitted in the course. **Marks will be deducted** if professionalism is lacking, e.g., homework is not identified, questions are answered out-of-order, papers are stapled out-of-order, units are not specified, etc.

Syllabus!

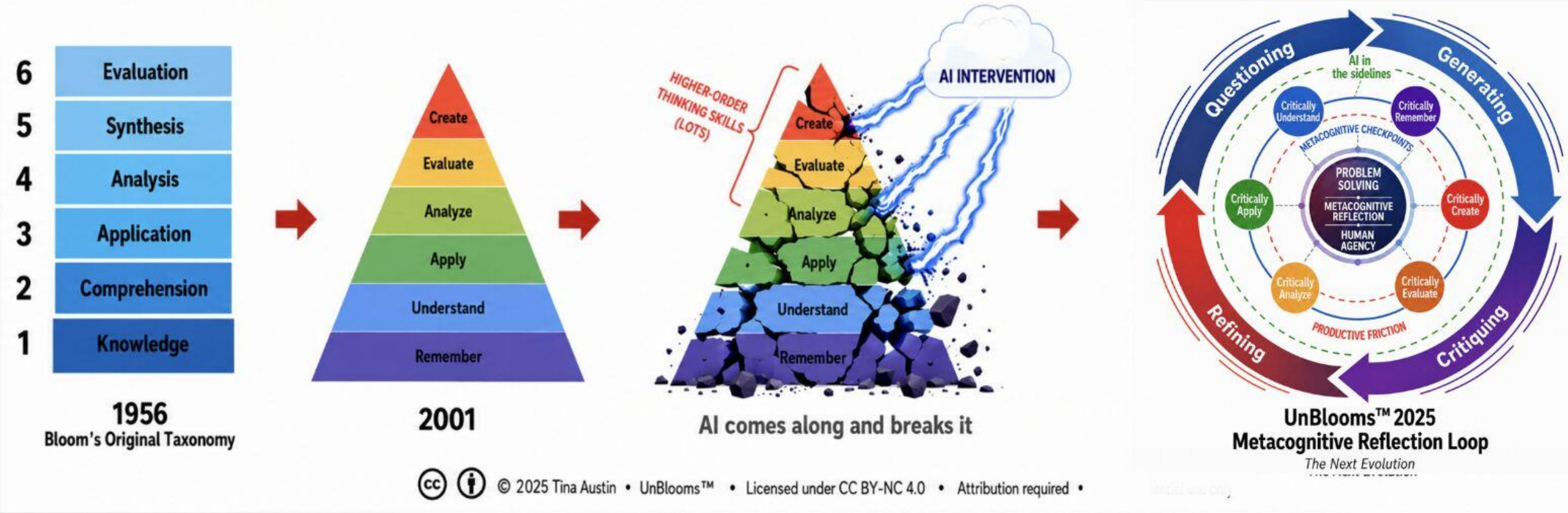
How to learn anything? (1/3)

Bloom's Taxonomy



How to learn anything? (2/3)

The Evolution of Bloom's Taxonomy to UnBlooms™



How to learn anything? (3/3)

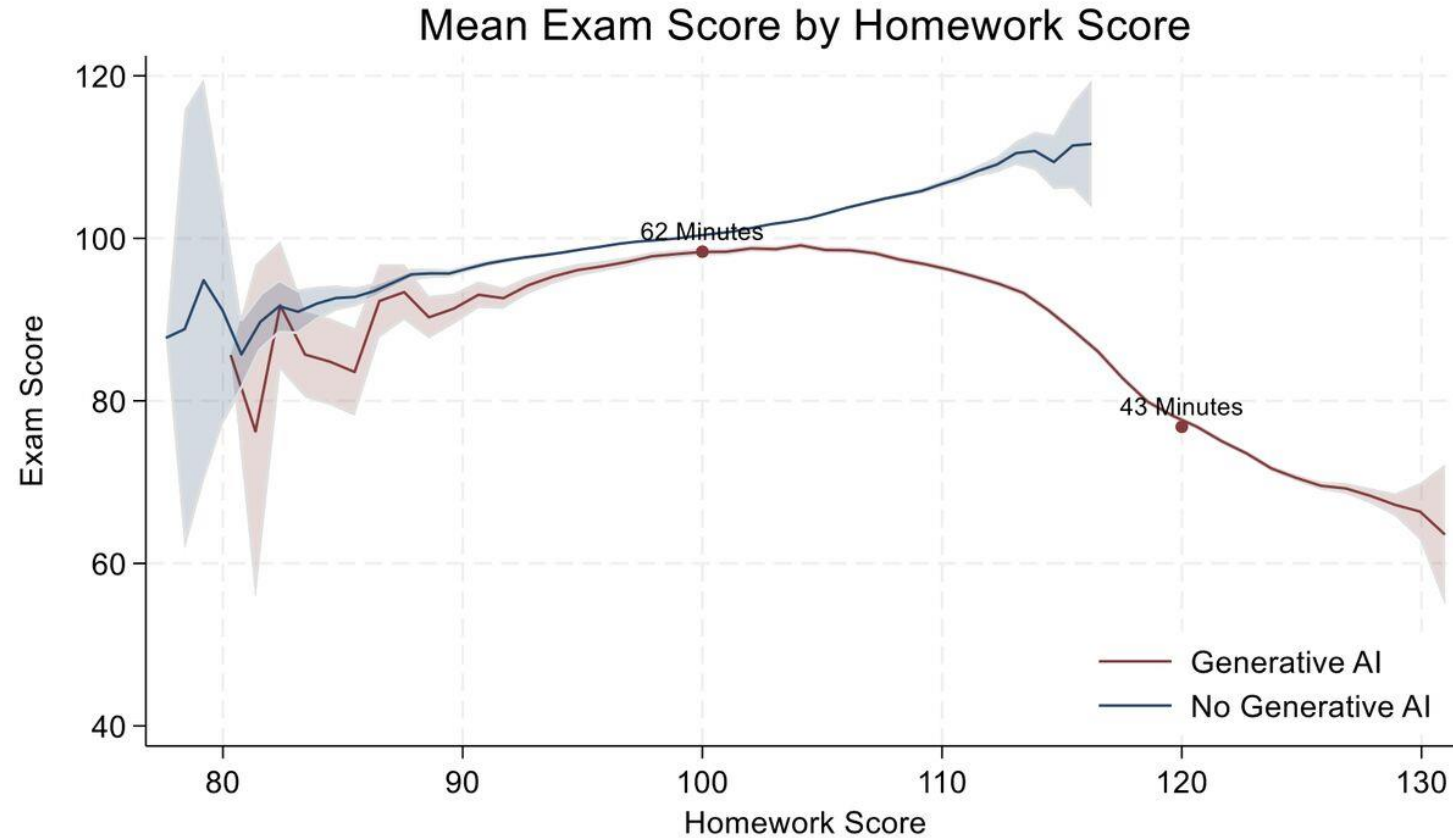
How does UnBlooms™ make a difference in learning?

Aspect	Traditional Bloom's	UnBlooms™
Starting point	Teacher delivers content → students absorb	Students generate initial understanding → test it against AI
AI's role	Threat to academic integrity OR ignored entirely	Partner for thinking <i>with</i> and <i>against</i>
Learning focus	Mastering enzyme facts and functions	Understanding <i>how we know</i> what we know about enzymes
Student thinking	Linear: memorize → apply → create	Recursive: predict → compare → revise → interrogate
Evidence of learning	Correct final answer/report	Visible reasoning process and metacognitive awareness
What's valued	Getting it right	Understanding why you were wrong and what that reveals
Failure is...	Something to hide	Data for learning

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What can you tell?

Figure A11. Homework and Exam Scores

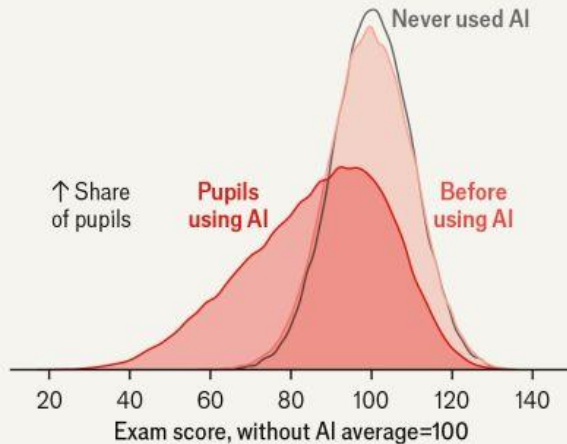


Note: The figure plots the smoothed mean and its 95% confidence intervals, by homework score, separately for students who have adopted generative AI and those who have not. The data also shows the average homework completion time for AI students with homework scores of 70 and 90. The data is at the student-month level and the sample summary statistics are shown in Panel A of Table 1.

What can you tell?

...but come exam time the technology proves less helpful

China*, Jan 2023-Jun 2025

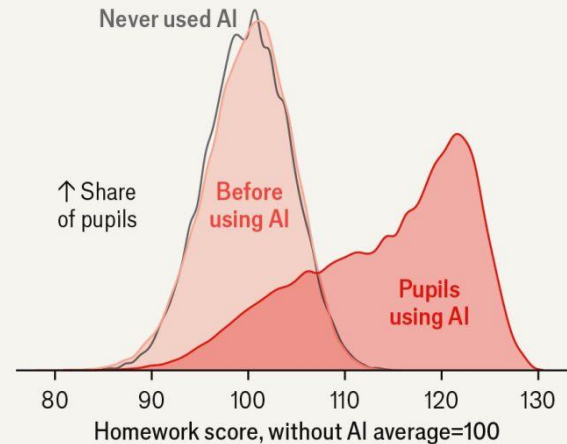


*26,811 pupils aged 12-18
Source: "The generative AI learning penalty",
by D. Strömberg et al., preprint June 2026

E

Pupils using AI have higher average homework scores...

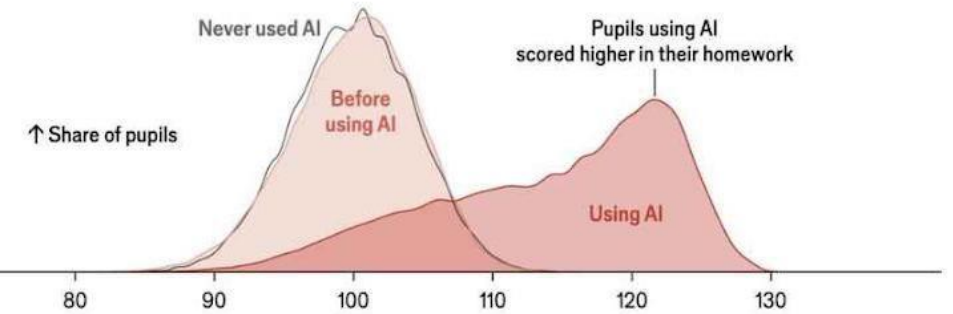
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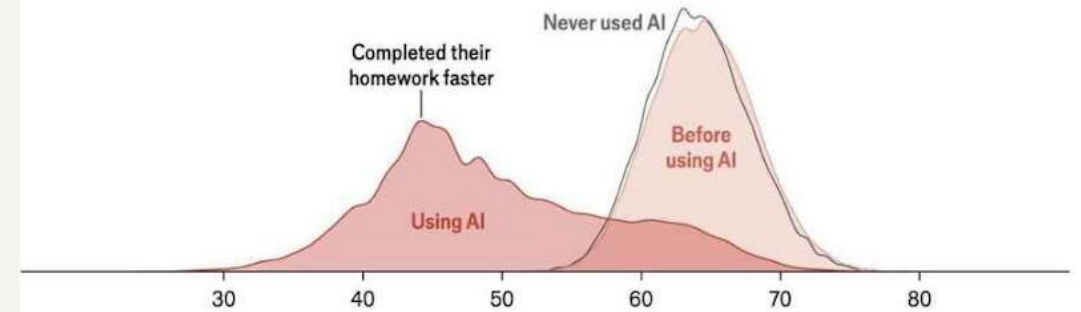
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Source: "The generative AI learning penalty",
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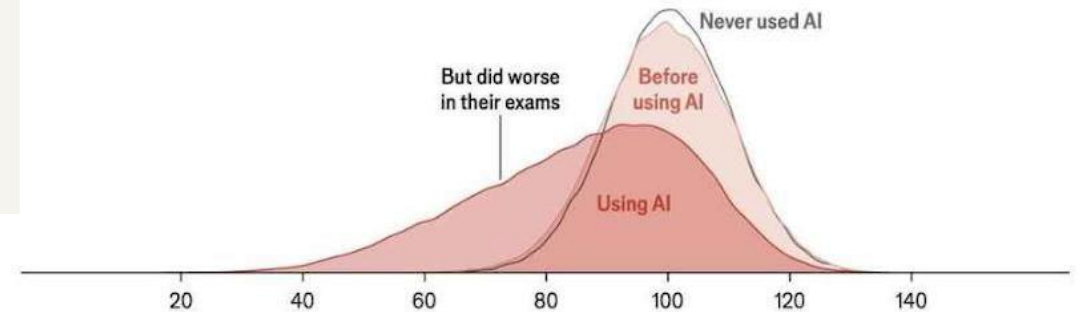
Homework score, average without AI=100



Homework completion time, minutes



Exam score, average without AI=100



Source: "The generative AI learning penalty: evidence from
Chinese secondary education", by D. Strömberg et al., preprint June 2026

*26,811 pupils aged 12-18

Expectations

Lectures

- 13 lectures total (2h each)
- Full-time and part-time students are expected to meet the same learning objectives
- Active engagement in class is expected (brownie cookie points for participation)

Assignments

- 4 assignments total (~2 weeks each)
- You are expected to practice coding on your own
- Strong self-learning ability is required (e.g., learning new tools or coding independently)
- I cannot help you debug your code

Readings

- Pre-readings based on sections of the NASA SE Handbook
- Post-readings: 1-2 journal or conference papers per week
- Expect to read ~30–40 pages per week
- Excellent learners take notes and self-organize knowledge across readings and lectures

Final Project

- Project proposal (Max 2 pages)
- 10-minute presentation
- Written final report

- System modeling is rapidly evolving
- Shifting from document-centric methods to Model-Based Systems Engineering (MBSE) for complex systems
- Imminent changes will place much greater emphasis on AI- and data-driven approaches, particularly for ill-defined systems and cyber-physical systems.

Machine Learning Systems Engineer
CentML • United States • via Indeed

Machine Learning Systems Engineer (1 Year Fixed Term)

Systems Research Engineer, Machine Learning Systems
Together AI • San Francisco, CA • via ZipRecruiter
\$ 160K-230K a year • Full-time • Health insurance

Artificial Intelligence Systems Engineer (DOD & Service Center)
The MITRE Corporation • Quantico, VA • via MITRE Jobs - MITRE Corporation
29 days ago • Full-time

Job highlights
Identified by Google from the original job post

Qualifications

- Strong background in machine learning, efficient inference for large language models
- Knowledge of ML/AI applications and as large language models and diffusion models and how they are used
- Knowledge of system performance and system architecture

2 more items(s)

Benefits

- We offer competitive compensation, benefits, as well as flexibility in terms of work schedule
- The US base salary range for this full equity + benefits
- Our salary ranges are determined by experience and location

Responsibilities

- As a Systems Research Engineer you will play a crucial role in researching and developing new machine learning systems at Together AI
- Working closely with the modeling, design and development teams to design large-scale distributed training inference engine that serves a diverse set of customers
- Your research skills will be vital in staying at the forefront of innovation in machine learning systems

Job highlights
Identified by Google from the original job post

Qualifications

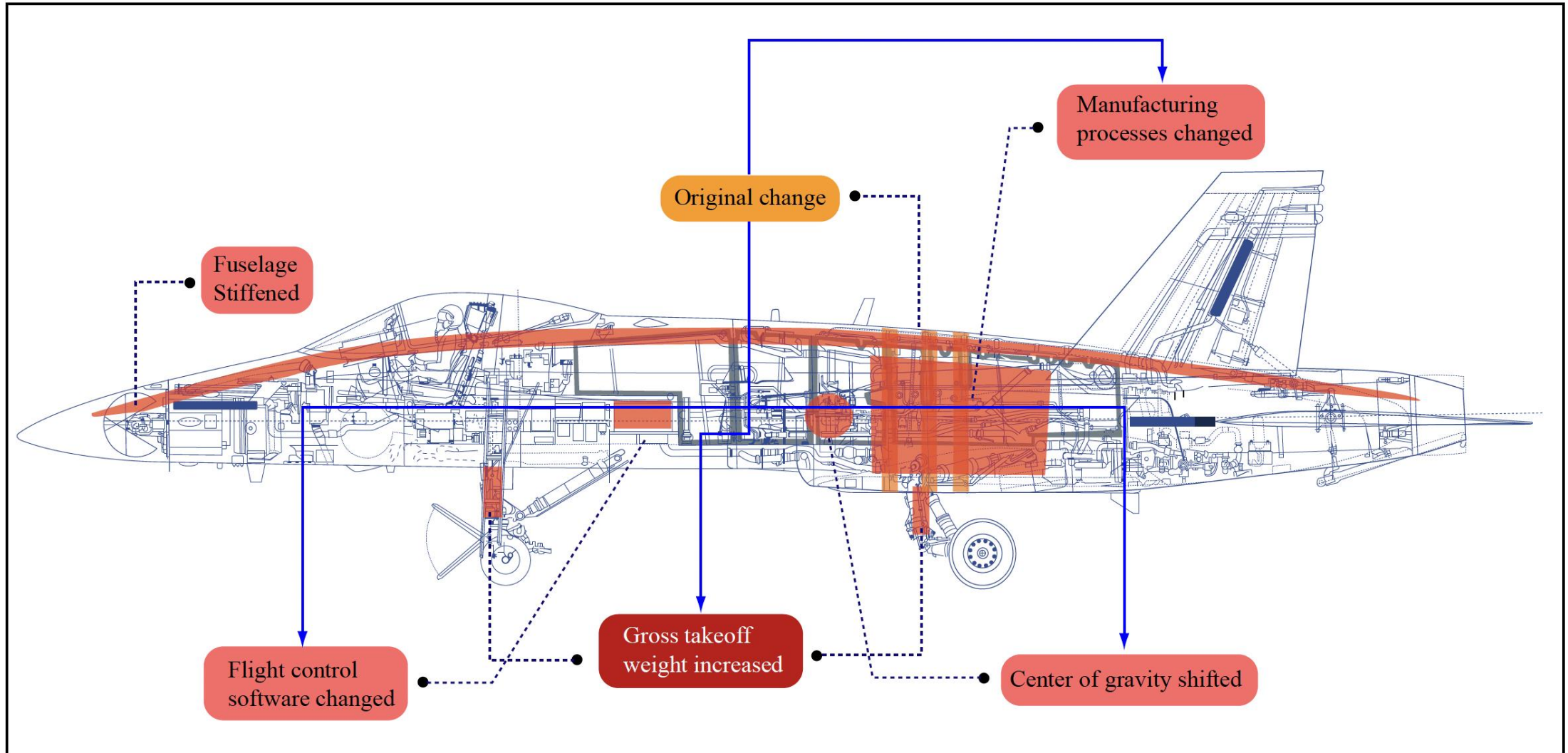
- The ideal candidate will have deep operational and domain knowledge of the Marine Corps meshed with an exquisite knowledge of current and future naval domain operational visions, doctrines, and tactical employments in the presence of US peer competitors
- The ideal candidate will be a champion for cultural and organizational transformation in data analytics, machine learning technologies, and the influence of artificial intelligence capabilities on society, industry, and defense
- Typically, interdisciplinary formal education background with at least a Bachelor of Science in a computer science field of study and 5 years of related experience; or 3 years and a Master's degree;

5 more items(s)

Benefits

- Salary compensation range and midpoint: \$142,000 - \$177,500 - \$213,000 Annual

F/A-18 Complex System Change



F/A-18 Complex System Change

- High-performance aircraft are very complex internally ... propulsion, avionics, structures ...
- Changing requirements can have ripple effects because everything is tightly coupled
 - It is difficult to predict the totality of system interactions ahead of time
- The “whole” system is much more than the air vehicle: logistics, training, incl. simulators etc..
- People matter a lot: contracts, culture, incentives

How would you define Systems Engineering?

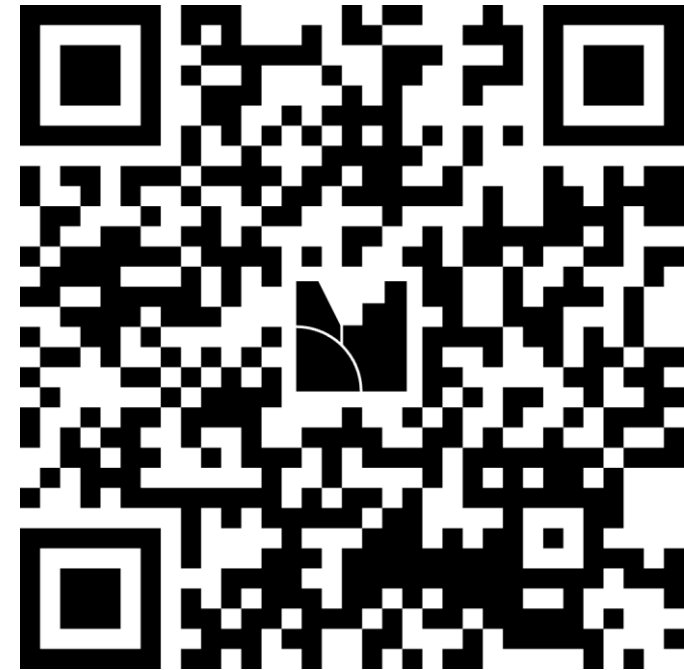
What is system modeling?

- Go to www.menti.com
- Enter code 7234 7177



How will system modeling shape future engineering practice?

- Go to www.menti.com
- Enter code 6980 9926



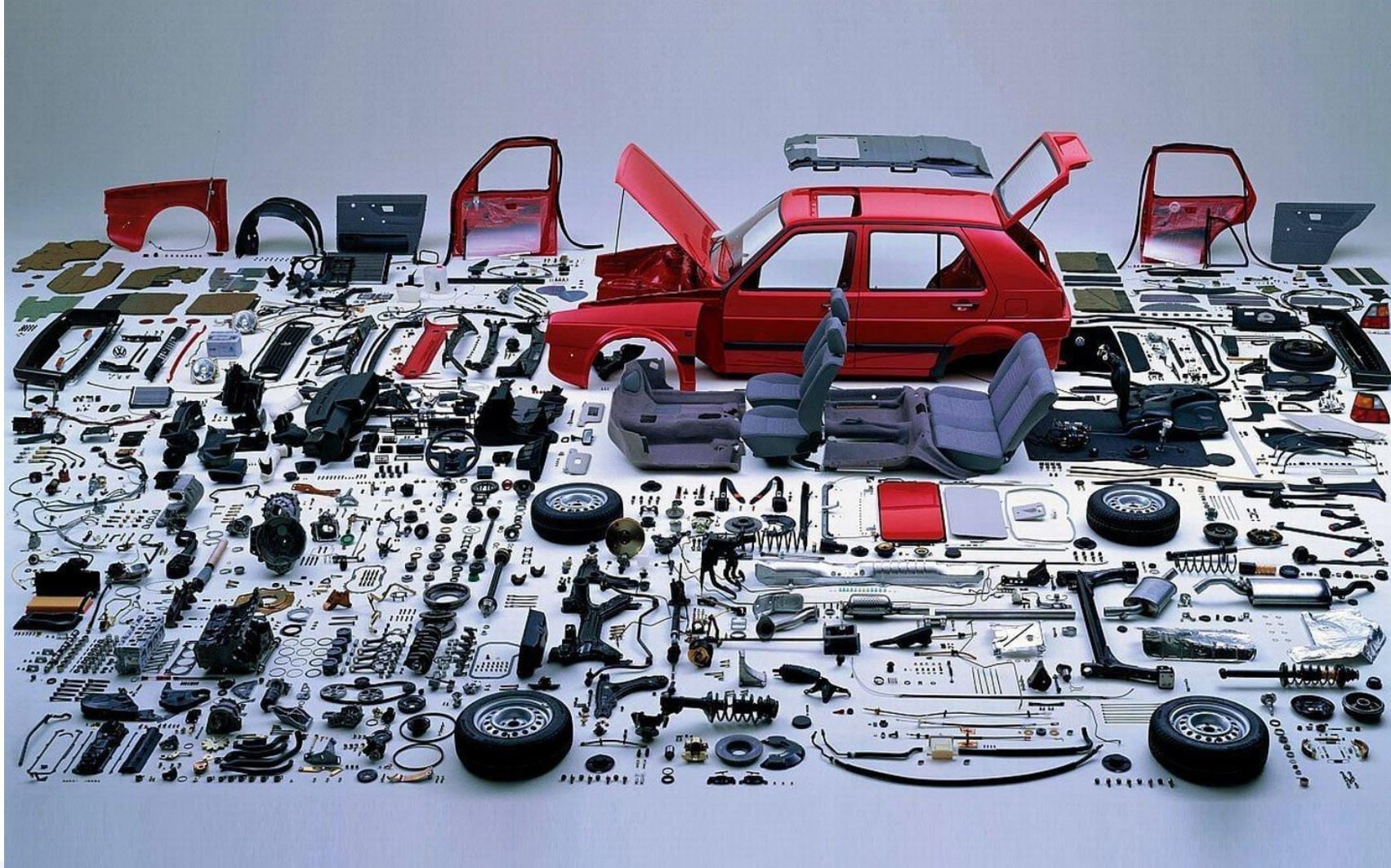
How would you define Systems Engineering?

- **Turn to your neighbor** and discuss for about **5 minutes**:
 - ☐ What is your definition of Systems Engineering?
 - ☐ Can you agree amongst yourselves?
 - ☐ What are the key elements of a definition?
- We will sample after about 5 minutes !

Motivation for this class

- Aerospace Systems deliver important functions to society ... air transportation, defense, sensing, exploration ...
- Complex “machines” with thousands of unique parts and potentially millions of interactions
 - Many aerospace systems require 6+ levels of decomposition to arrive at indivisible parts that cannot be taken “a-part”
- Humans play an important role as designers, operators, beneficiaries, maintainers
- Best Practices have emerged since the 1960’s and are continuously evolving ... documented in standards/handbooks
- Limitations of “traditional” SE
 - **System safety** ... recent SpaceX Falcon 9 launch failure
 - **Typical program cost and schedule overruns** ... Boeing Dreamliner 787 delays ...
- Systems Engineering is also penetrating in other industries
 - Automobiles, Software, Medical Devices

Why do we need system decomposition?



[Image Source:](#)

System Complexity

Assume 7-tree [Miller 1956]
<http://www.musanim.com/miller1956/>

- How many levels in drawing tree?

$$\#levels = \frac{\log(\#parts)}{\log(7)}$$

		#parts	#levels
•	Screwdriver (B&D)	3	1
•	Roller Blades (Bauer)	30	2
•	Inkjet Printer (HP)	300	3
•	Copy Machine (Xerox)	2,000	4
•	Automobile (GM)	10,000	5
•	Airliner (Boeing)	100,000	6

Simple



complex

Source: Ulrich, K.T., Eppinger S.D. , Product Design and Development
Second Edition, McGraw Hill, 2nd edition, 2000, Exhibit 1-3

Concept Question:

- How many levels of decomposition (depth of drawing tree) do we need to describe the car shown in the previous picture?
- 1
- 2
- 3
- 4
- 5
- 6
- >6
- This question does not make sense to me

Learning Objectives

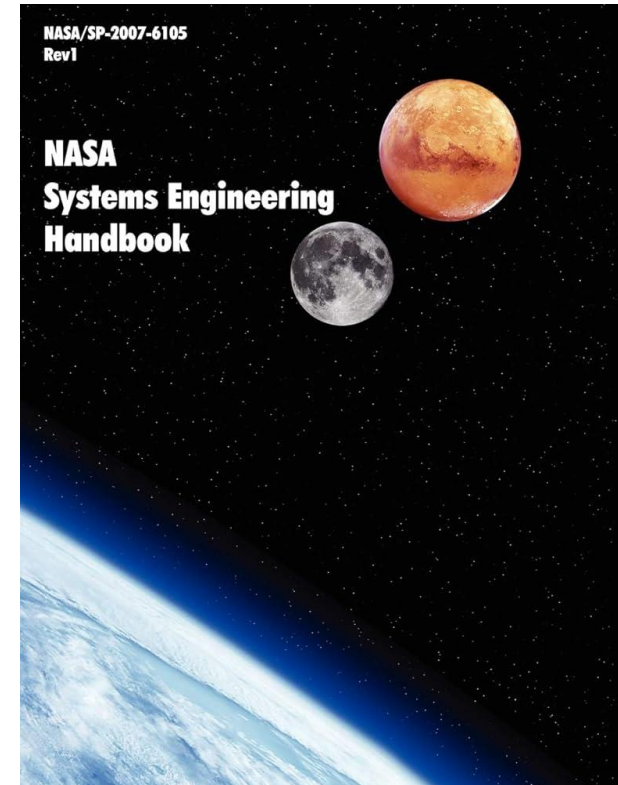
Participants in this class will be able to ...

- SE1: Describe the most important **Systems Engineering standards and best practices** as well as newly emerging approaches [1]
- SE2: Structure the **key steps in the systems engineering process** starting with stakeholder analysis and ending with transitioning systems to operations
- SE3: Analyze the **important role of humans** as beneficiaries, designers, operators and maintainers of aerospace and other systems
- SE4: Characterize the **limitations of the way that current systems engineering is** practiced in terms of dealing with complexity, lifecycle uncertainty and other factors
- SE5: **Apply some of the fundamental methods and tools** of systems engineering to a ‘simple’ cyber-electro-mechanical system as a stepping stone to more complex and real-world projects

[1] Our main “textbook” for the class will be the NASA Systems Engineering Handbook, NASA/TP-2007-6105, Rev 1.

NASA/SP-2007- 6105

- “Bible” for Systems Engineering at NASA
- Makes The Bridge From “Typical” Guidance Back To NASA Systems Engineering Process (NPR 7123.1)
 - Guidance From Practitioners
 - Written by practitioners for practitioners
 - “How” Vs “What”
- Updates The Guidance from SP-6105 (basic)
 - Updates The Practice/Methodology from 1995
- Provides Top-level Guidance for Systems Engineering Best Practices; It Is Not Intended In Any Way To Be A Directive
- Adds Additional Special Topics
 - Tools
 - NEPA
 - Human Factors



Learning Objectives

- Introductions
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A bit of SE History

- Systems Engineering has been informally practiced since antiquity
 - Great Wall of China, Egyptian Pyramids, Roman Aqueducts
 - Mainly a “workforce” problem to build large infrastructures
- The term “**Systems Engineering**” can be traced back to Bell Labs (1940s)
 - https://en.wikipedia.org/wiki/Bell_Labs
 - Beginning of new methods to better handle complexity
- Formal Systems Engineering really started after WWII
 - 1950’s and 1960s: Cold War, Apollo Lunar Program, ICBMs etc...
 - Complex Engineering Systems: Air Traffic Control, High Speed Rail, Nuclear ..
 - Mainly (paper) document-based: requirements, specifications, test plans etc...
- Early Pioneers
 - Arthur D. Hall, Kelly Johnson, Simon Ramo, Eberhard Rechtin, Andrew Sage, Margaret Hamilton, and others
- 1995 Founding of International Council for Systems Engineering (INCOSE)
- Since ~2000: Development of new Model-Based-Systems-Engineering (MBSE). Need to accelerate SE and better handle complexity



Simon Ramo

Some Definitions of SE

- "System engineering is a robust approach to the design, creation, and operation of systems. In simple terms, the approach consists of identification and quantification of system goals, creation of alternative system design concepts, performance of design trades, selection and implementation of the best design, verification that the design is properly built and integrated, and post-implementation assessment of how well the system meets (or met) the goals."- ***NASA Systems Engineering Handbook, 1995.***
- "An interdisciplinary approach and means to enable the realization of successful systems"- ***INCOSE handbook, 2004***
- More recently the scope of SE has broadened:
 - Design of Enterprises, Infrastructure Networks etc...
- ***AI4SE or SE4AI:***
 - How do we leverage AI-enabled capabilities to improve SE execution? (*AI4SE*)
 - How do we execute SE activities on AI-enabled systems? (*SE4AI*)

